Rates of Change & Concavity

ANSWER KEY

Section 1: Average Rate of Change (AROC)

1 Calculate the average rate of change of $f(x) = 2x^2 - 3x + 1$ on the interval $[1, 4]$.

a. 7 ✓

b. 9

c. 21

d. 6

AROC $= \frac{f(4) - f(1)}{4 - 1}$. $f(4) = 2(16) - 12 + 1 = 21$. $f(1) = 2 - 3 + 1 = 0$. AROC $= \frac{21 - 0}{3} = 7$.

2 A vehicle travels such that its distance in meters is given by $s(t) = 3t^2 + 10t$. What is the average velocity from $t = 2$ to $t = 5$ seconds?

a. 31 m/s ✓

b. 25 m/s

c. 62 m/s

d. 15 m/s

$s(5) = 3(25) + 50 = 125$. $s(2) = 3(4) + 20 = 32$. Average velocity $= \frac{125 - 32}{5 - 2} = \frac{93}{3} = 31$ m/s.

3 For a linear function $g(x) = mx + b$, what can be said about the average rate of change on any interval $[a, b]$?

a. **It is constant and always equal to** m ✓

b. It depends on the midpoint $\frac{a + b}{2}$

c. It is zero if $b > 0$

d. It equals b

Linear functions have a constant rate of change equal to the slope m over any interval.

Section 2: Increasing/Decreasing and Changing Rates

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4 If a function f is increasing at an increasing rate on an interval, what describes its graph?

a. **The curve goes upward and is concave up ✓**

b. The curve goes upward and is concave down

c. The curve goes downward and is concave up

d. The curve goes downward and is concave down

Increasing means the values are rising (upward). At an increasing rate means the slope is getting steeper, which corresponds to concave up geometry.

5 A function g has average rates of change over consecutive unit intervals: on $[0, 1]$ AROC is 8, on $[1, 2]$ AROC is 5, on $[2, 3]$ AROC is 2. Which model best characterizes g?

a. g **is increasing at a decreasing rate (concave down) ✓**

b. g is increasing at an increasing rate (concave up)

c. g is decreasing at a constant rate

d. g is decreasing at an increasing rate

The AROC values are positive $(8, 5, 2)$, so g is increasing. However, the rates are decreasing $(8 > 5 > 2$), which signifies concave down behavior.

6 If $h(x)$ is decreasing at an increasing rate on an interval, what happens to the slope of secant lines?

a. **The slopes become less negative (closer to zero) ✓**

b. The slopes become more negative (steeper downward)

c. The slopes remain constant

d. The slopes switch from positive to negative

Decreasing means rates are negative. At an increasing rate means the rate values are increasing (e.g. from -10 to -4 to -1), making the slope less negative and flattening out (concave up).

Section 3: Concavity & Inflection Points

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7 Which of the following points represents an inflection point on the graph of $y = f(x)$?

a. **A point where the concavity changes from concave up to concave down (or vice versa)** ✓

b. A point where the function reaches an absolute maximum

c. A point where the average rate of change is zero

d. A point where the graph crosses the x-axis

An inflection point is defined as a point on the curve where the concavity of the graph changes from upward to downward or from downward to upward.

8 On the interval $(-\infty, 0)$, f is concave up; on $(0, \infty)$, f is concave down. If $f(0) = 4$, which statement is true?

a. $(0, 4)$ **is an inflection point of** f ✓

b. $(0, 4)$ is a local minimum of f

c. $(0, 4)$ is a local maximum of f

d. $f'(0)$ must equal 0

Because the concavity changes at $x = 0$ and $(0, 4)$ lies on the continuous curve, $(0, 4)$ is an inflection point.

9 The graph of $y = x^3 - 3x$ has an inflection point at which coordinate?

a. $(0, 0)$ ✓

b. $(1, -2)$

c. $(-1, 2)$

d. $(0, -3)$

For a cubic $y = ax^3 + bx^2 + cx + d$, the inflection point occurs at $x = -b/(3a) = 0$. At $x = 0$, $y = 0$, giving $(0, 0)$.

Section 4: Contextual Rates of Change Applications

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10 Water is poured into a conical cup at a constant rate. What is the concavity of the depth of water $h(t)$ as a function of time?

a. **Concave down, because the radius increases with height so height increases at a decreasing rate ✓**

b. Concave up, because height increases at an increasing rate

c. Linear, because the flow rate is constant

d. Constant concavity of zero

Because the cup widens as it fills, each additional volume unit adds progressively less height. Thus the rate of change of height decreases over time, giving a concave down graph.

11 The table shows the temperature $T(t)$ in $^\circ$F of cooling coffee: $T(0) = 180$, $T(5) = 150$, $T(10) = 126$. What is the average rate of cooling from $t = 0$ to $t = 10$ minutes?

a. -5.4°F/min ✓

b. -6.0°F/min

c. -4.8°F/min

d. -2.7°F/min

$\text{AROC} = \frac{T(10) - T(0)}{10 - 0} = \frac{126 - 180}{10} = \frac{-54}{10} = -5.4^\circ\text{F/min}.$

12 If the average rate of change of f on $[2, 2 + h]$ approaches 12 as $h \to 0$, what does 12 represent?

a. **The instantaneous rate of change of f at $x = 2$ ✓**

b. The average rate of change of f on $[0, 2]$

c. The value of $f(2)$

d. The maximum value of f

The limit of the average rate of change as the interval width $h \to 0$ represents the instantaneous rate of change (derivative) at $x = 2$.

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